

# Exam Question: Flood-Warning Sensor Model

Suggested time: 30 minutes | Total: 20 marks | Show all reasoning and state assumptions

Context. A river-monitoring station classifies each hour as either high-water (H) or normal (N). Historical records suggest  $P(H)=0.08$ . The sensor raises an alarm A with sensitivity  $P(A|H)=0.92$  and false-alarm probability  $P(A|N)=0.06$ .

## 1. Alarm probability and diagnosis (7 marks)

- (a) Use the law of total probability to calculate  $P(A)$ . [3]
- (b) An alarm is observed. Calculate  $P(H|A)$ , and explain in one sentence why this value is not equal to the sensor sensitivity. [4]

## 2. A discrete random variable (5 marks)

Let  $X$  be the number of alarms in the next 4 hours. For this part only, assume the four hours are independent and each has the same marginal alarm probability found in Question 1(a).

- (a) State the probability model for  $X$ , including all parameters. [2]
- (b) Calculate  $P(X=1)$ . [2]
- (c) State  $E[X]$ . [1]

## 3. Continuous measurement error (5 marks)

When the station is calibrated, its signed measurement error  $E$  (in cm) is modeled as uniformly distributed on  $[-2,2]$ .

- (a) Write the probability density function  $f_E(e)$ , including its support. [2]
- (b) Calculate  $P(|E| \leq 0.5)$ . [2]
- (c) Find the median of  $E$ . [1]

## 4. Modelling judgement (3 marks)

The station manager proposes using the independence assumption from Question 2 during a multi-day storm. Identify one reason this assumption may fail and describe one practical consequence for the model results. [3]

Formula reminders.  $P(B|A)=P(A|B)P(B)/P(A)$ . For  $X \sim \text{Binomial}(n,p)$ ,  $P(X=k)=C(n,k)p^k(1-p)^{n-k}$  and  $E[X]=np$ .

Instructions. Numerical answers should be given to at least three significant figures. Unsupported calculator output may receive reduced credit. Clearly label events and random variables.